

Joint modeling of social genetic effects in mono- and pluri-specific groups: case study in intercrops

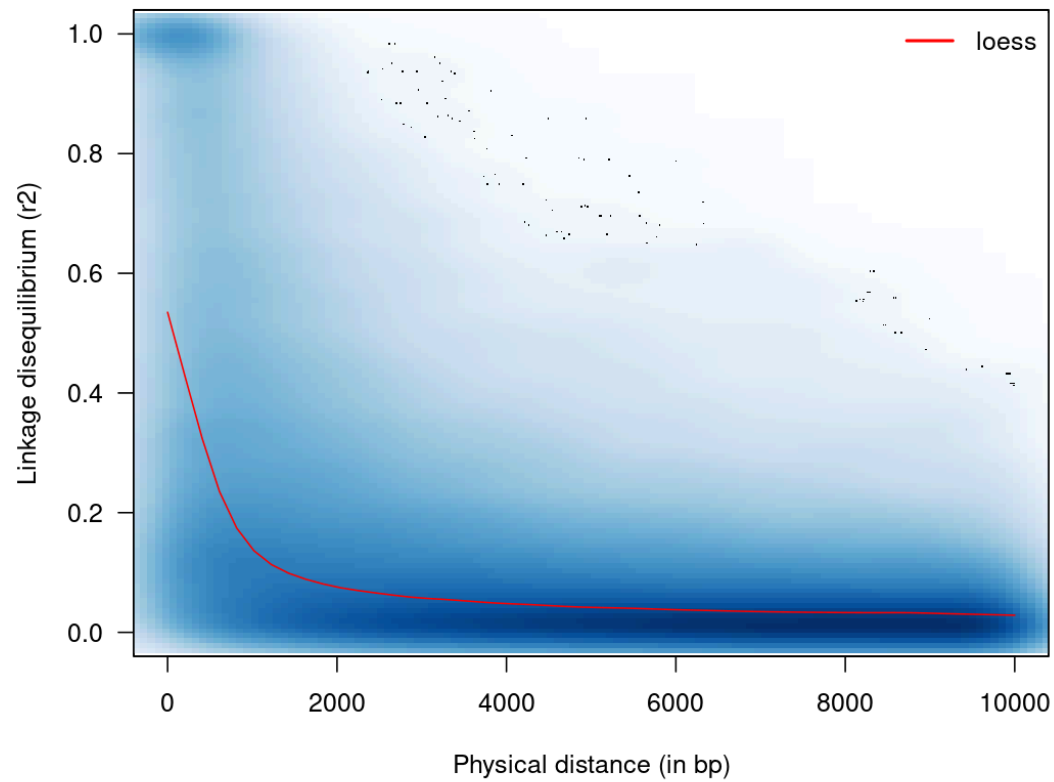
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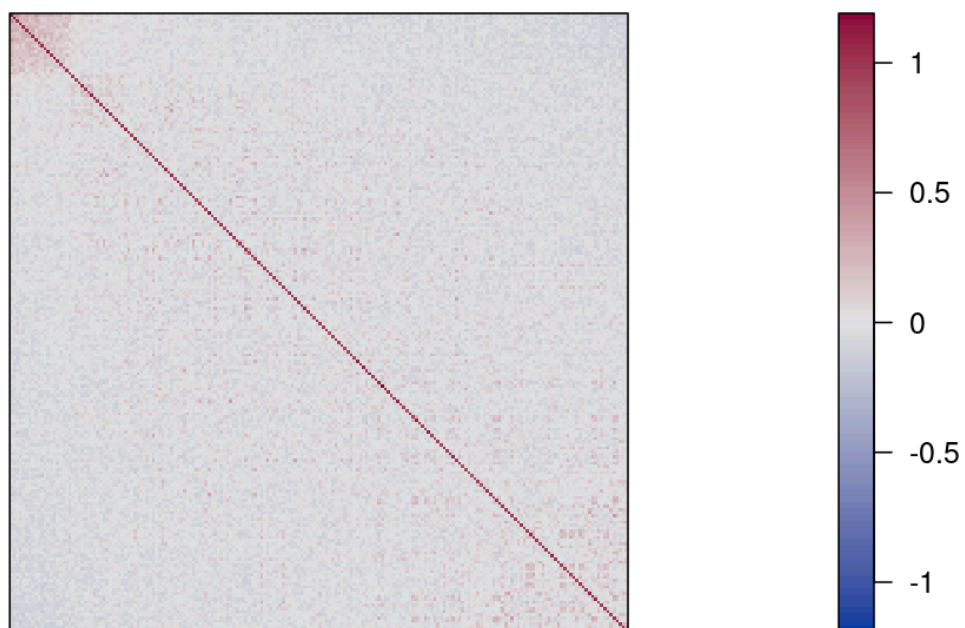
Supplementary File 2

Supplementary Figure 1: Linkage disequilibrium between SNPs in the simulated data set

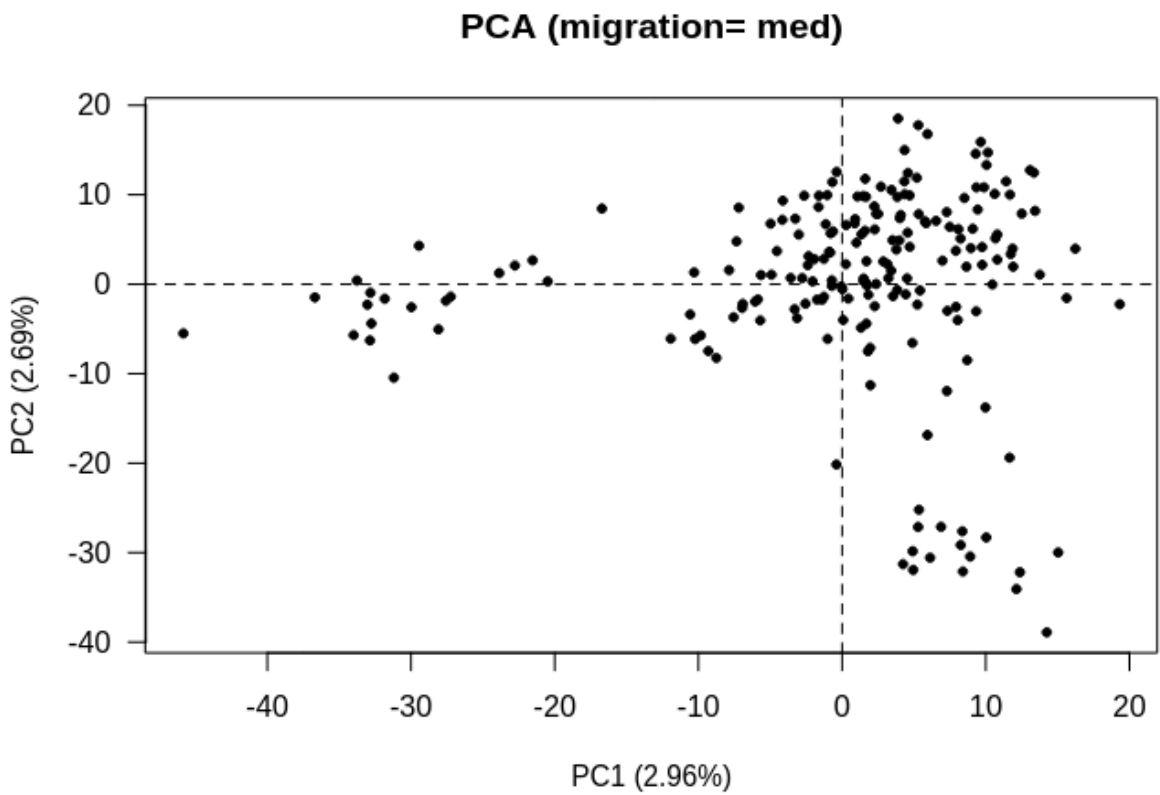


Supplementary Figure 2: Genomic relationship matrix in the simulated data set

Additive genetic relationships (migration=med)



Supplementary Figure 3: Genetic structure among genotypes in the simulated data set



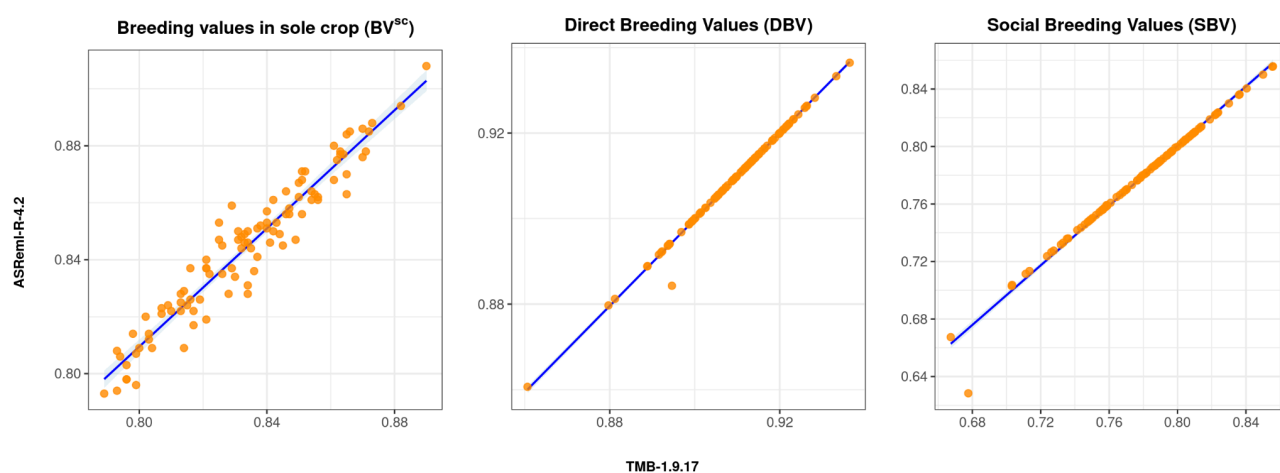
Supplementary Table 1:Computational efficiency of ASReml and TMB

Model	Designs	TMB (in seconds)	ASReml-R (in seconds)
<i>Univariate</i>	sole_only	<i>Auto. diff.: 37.00</i> <i>Nll^l minimisation (R): 4.30</i> <i>Total: 41.30</i>	<i>Total: 0.10</i>
<i>Multivariate</i>	inter_only	<i>Auto. diff. : 30.38</i> <i>Nll^l minimisation (R): 27.15</i> <i>Total: 57.53</i>	<i>Total: 0.27</i>
	sole_inter_50	<i>Auto. diff.: 16.05</i> <i>Nll^l minimisation (R): 166.01</i> <i>Total: 172.06</i>	<i>NA</i>

^l: negative log-likelihood

Supplementary Table 1: The computational efficiencies of ASReml-R and TMB were assessed for each design. The values in the table correspond to averages over 100 replicates, simulated with $\text{var}(\text{SBV}) = 20\% \text{var}(\text{DBV})$, $\text{var}(\text{SIGV}) = 10\% \text{var}(\text{DBV})$ and $\text{cor}(\text{DBV}, \text{SBV}) = -0.8$.

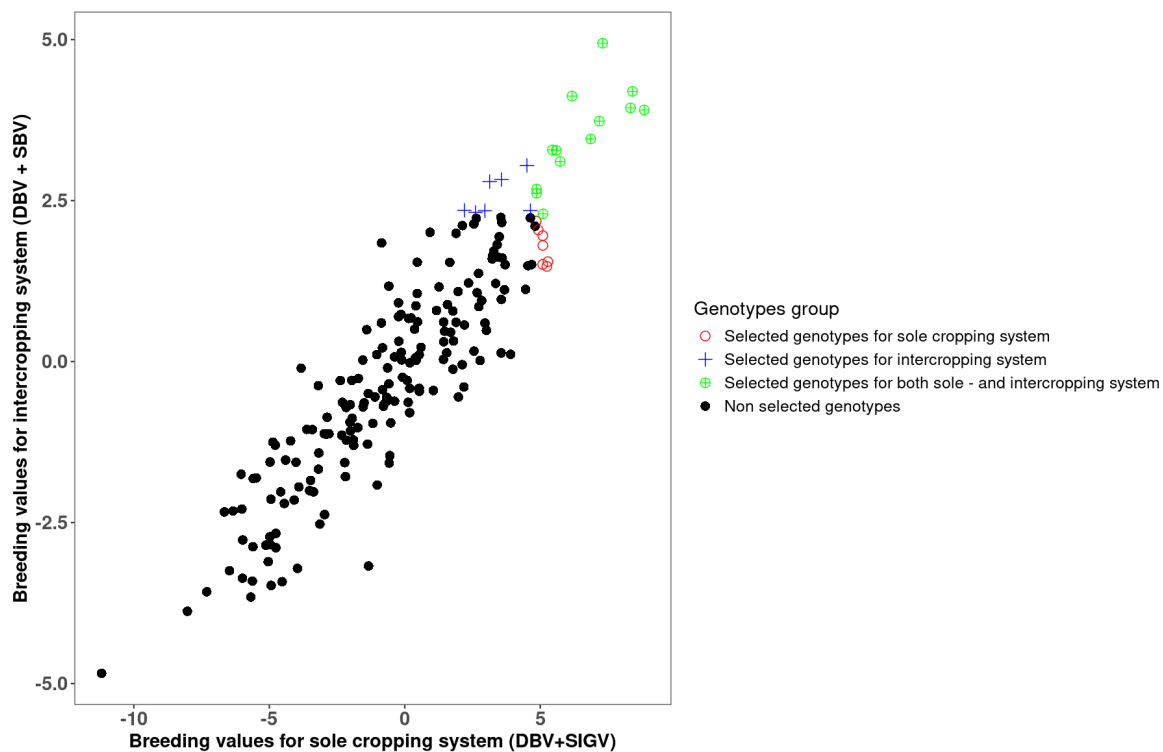
Supplementary Figure 4: Estimation accuracy of ASReml and TMB



Supplementary Figure 4: Comparison of estimated breeding values obtained using ASReml-R 4.2 and TMB 1.9.17. The figure presents breeding values in solecropping (BV^{SC}) estimated from the sole_only design, the direct breeding values (DBV) and social breeding values (SBV) from the inter_only design. Each panel displays a scatter plot comparing estimates from both methods across replicates, with individual points representing data distributions.

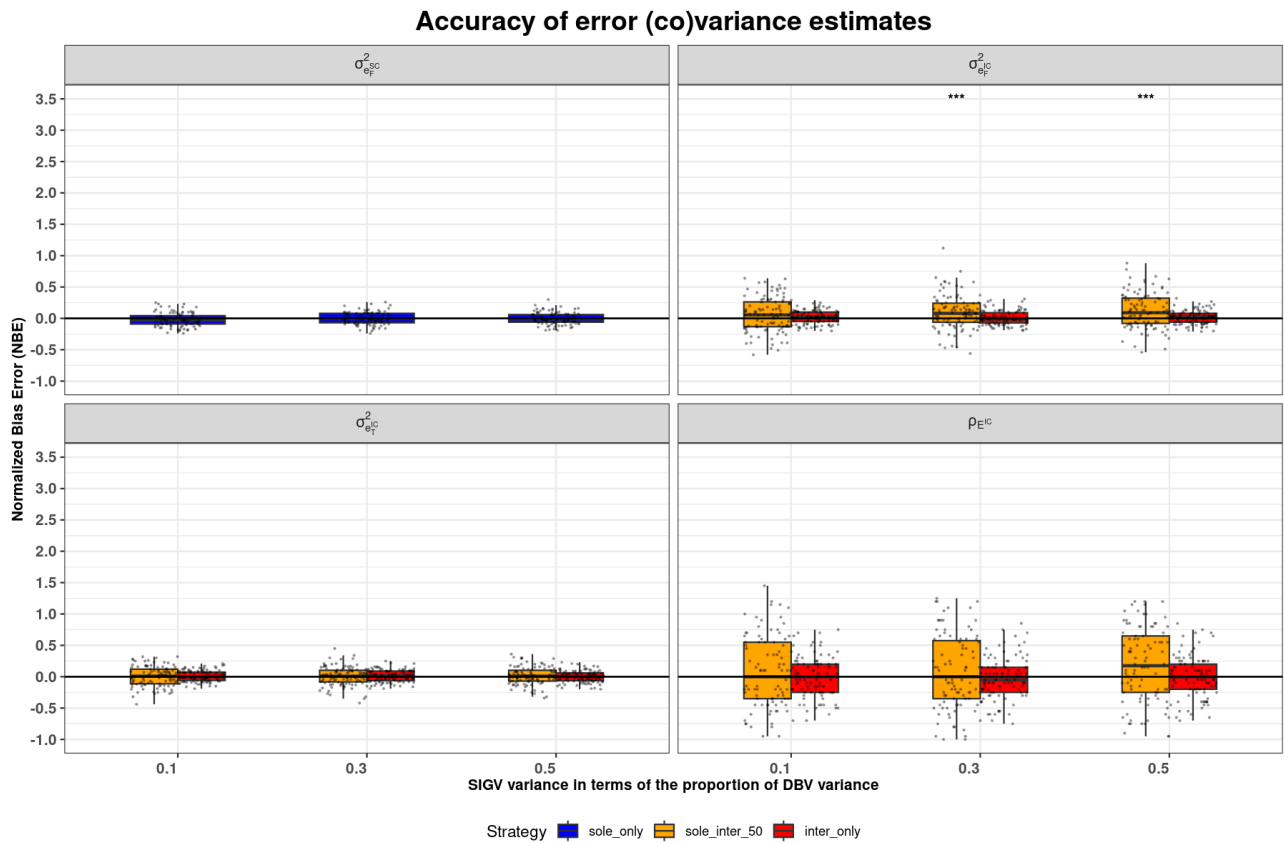
TMB and ASReml-R exhibited an almost perfect correlation for the estimation of direct breeding values (DBV: $R_{adj2} = 0.992$, $RSE=0.001$) and social breeding values (SBV: $R_{adj2} = 0.984$, $RSE=0.005$), but a slightly less perfect correlation for the breeding values in solecrop (BV^{SC} : $R_{adj2} = 0.924$, $RSE=0.007$).

Supplementary Figure 5: Selection of genotypes in a bivariate distribution



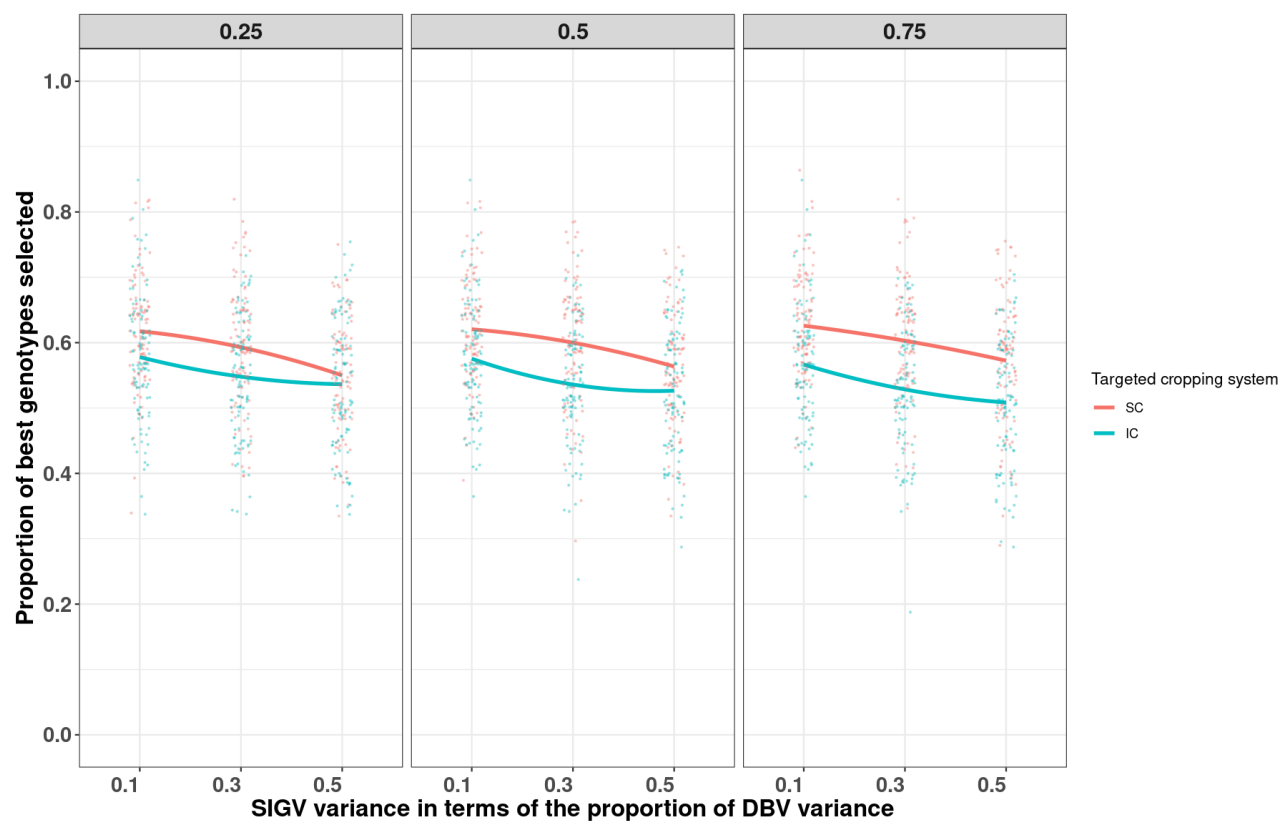
Supplementary Figure 5: Selection of genotypes in a bivariate distribution on one simulated dataset with the selection index for the sole cropping system ($BV^{SC} = DBV + SIGV$) on the x-axis and the selection index for the intercropping system ($BV^{IC} = DBV + SBV$) on the y-axis. Simulation parameters are defined as follow: sole_inter_50; $\text{var}(SBV) = 20\% \text{var}(DBV)$; $\text{var}(SIGV) = 50\% \text{var}(DBV)$, $\text{cor}(DBV, SBV) = -0.8$. A selection proportion of 10% was first applied independently for the sole cropping system and then for the intercropping system. Genotypes selected for both systems correspond to the common genotypes between them.

Supplementary Figure 6: Accuracy of error (co)variance estimates



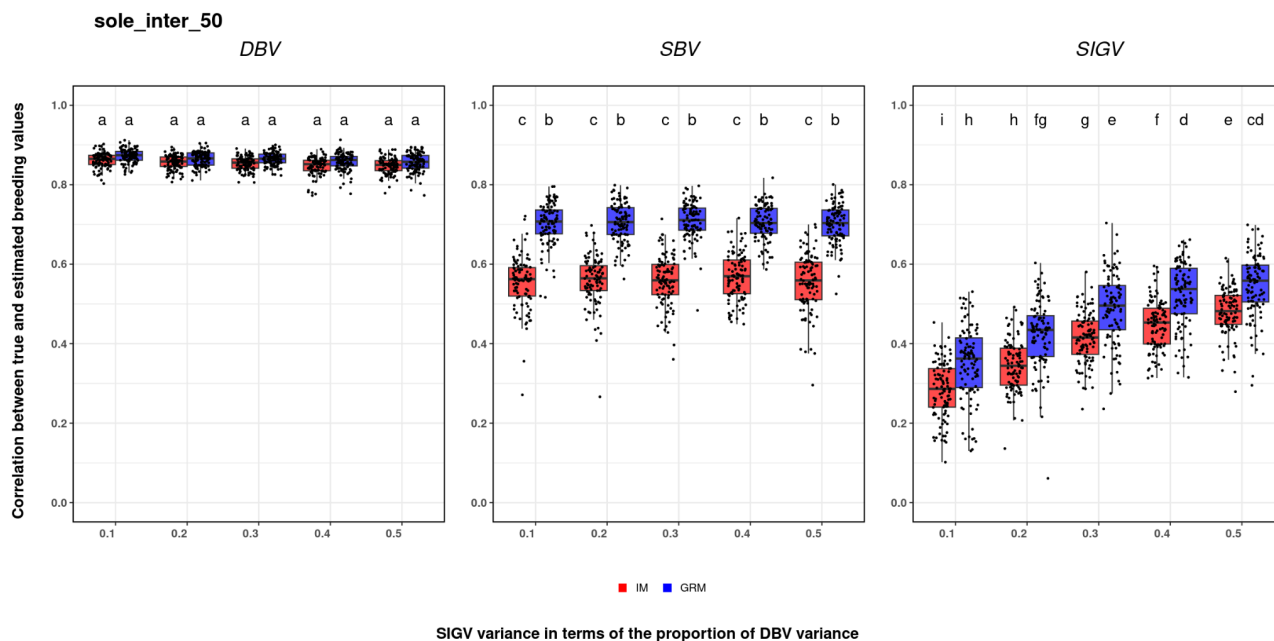
Supplementary Figure 6: Estimation accuracy (normalized biased error on the y-axis, one dot per simulation) of error (co)variances (F= focal species; T= tester species), under varying levels of SIGV variance (x-axis) ($\text{var}(\text{SBV}) = 20\% \text{var}(\text{DBV})$). “*” above each box indicate whether the mean normalized biased error is significantly different from zero based on a one-sample t-test: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Non-significant results are left unlabeled.

Supplementary Figure 7: Influence of the social intragenotypic variance on genotypes selection



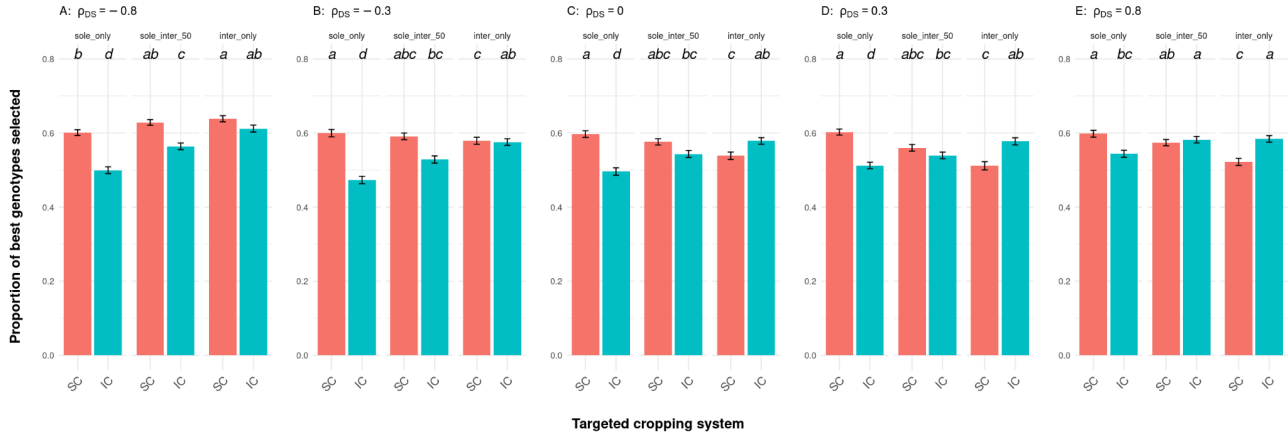
Supplementary Figure 7: Proportion of top-performing genotypes selected for sole cropping and intercropping across varying levels of social intragenotypic variance (x-axis), the data being simulated with $\text{var}(\text{SBV}) = 20\% \text{ var}(\text{DBV})$. Results are shown for $w_{sole} = 0.25, 0.50$, and 0.75 .

Supplementary Figure 8: Accuracy with or without genetic relationships over a range of SIGV variance



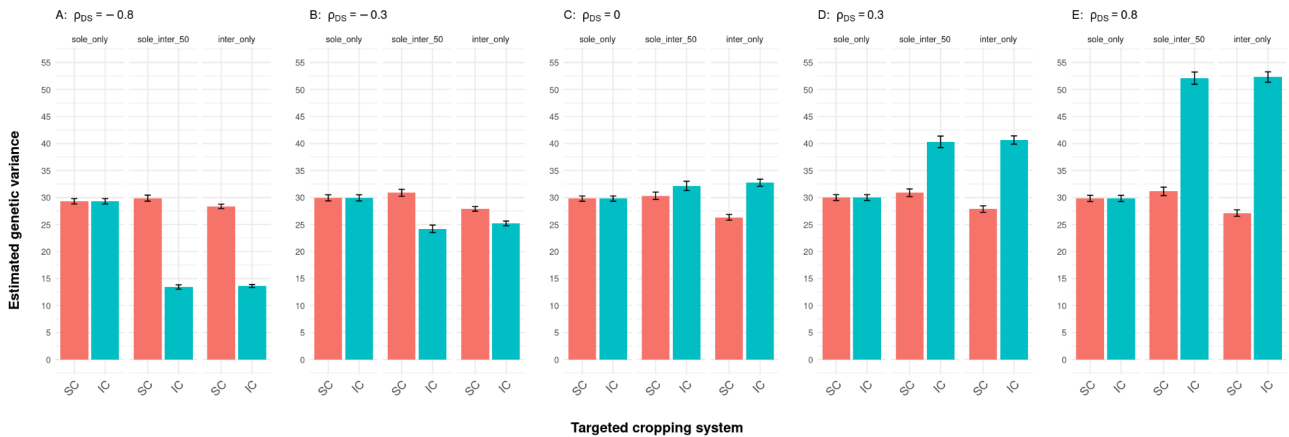
Supplementary Figure 8: Pearson correlations between true and estimated breeding values (y-axis) across varying levels of SIGV variance ($\text{var}(\text{SBV}) = 20\% \text{ var}(\text{DBV})$). Estimates obtained with an identity matrix (IM) are shown in red, whereas estimates with a genomic relationship matrix (GRM) are shown in blue. Groups that do not share a common letter differ significantly according to Tukey's HSD test at the 5% significance level.

Supplementary Figure 9: Influence of the correlation between DBV and SBV on genotype selection



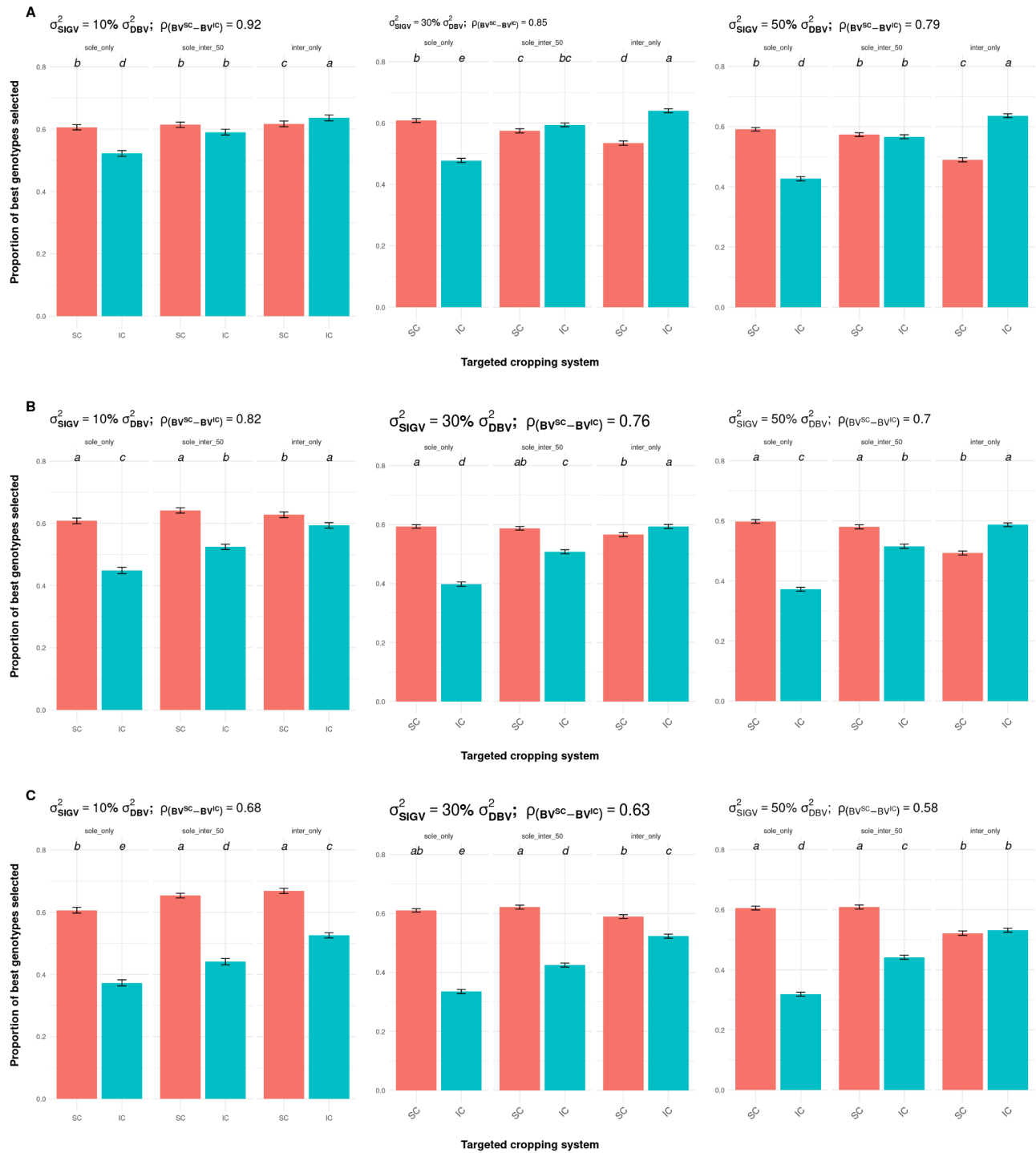
Supplementary Figure 9: Proportion of top-performing genotypes (y-axis) selected for a targeted cropping system (x-axis; sole cropping, SC, or intercropping, IC), based on their breeding values estimated from three experimental designs (sole_only, sole_inter_50 and inter_only). Results are shown across varying signs and magnitudes of $\text{cor}(\text{DBV}, \text{SBV})$: -0.8 (A), -0.3 (B), 0 (C), 0.3 (D) and 0.8 (E). Other (co)variance parameters were fixed as in figure 6: $\text{var}(\text{SBV}) = 20\% \text{ var}(\text{DBV})$ and $\text{var}(\text{SIGV}) = 10\% \text{ var}(\text{DBV})$. Error bars represent standard errors of the mean across replicates. Tukey's HSD test was performed separately within each level of $\text{cor}(\text{DBV}, \text{SBV})$; groups that do not share a common letter differ significantly ($\alpha = 0.05$).

Supplementary Figure 10: Influence of the correlation between DBV and SBV on total genetic variance



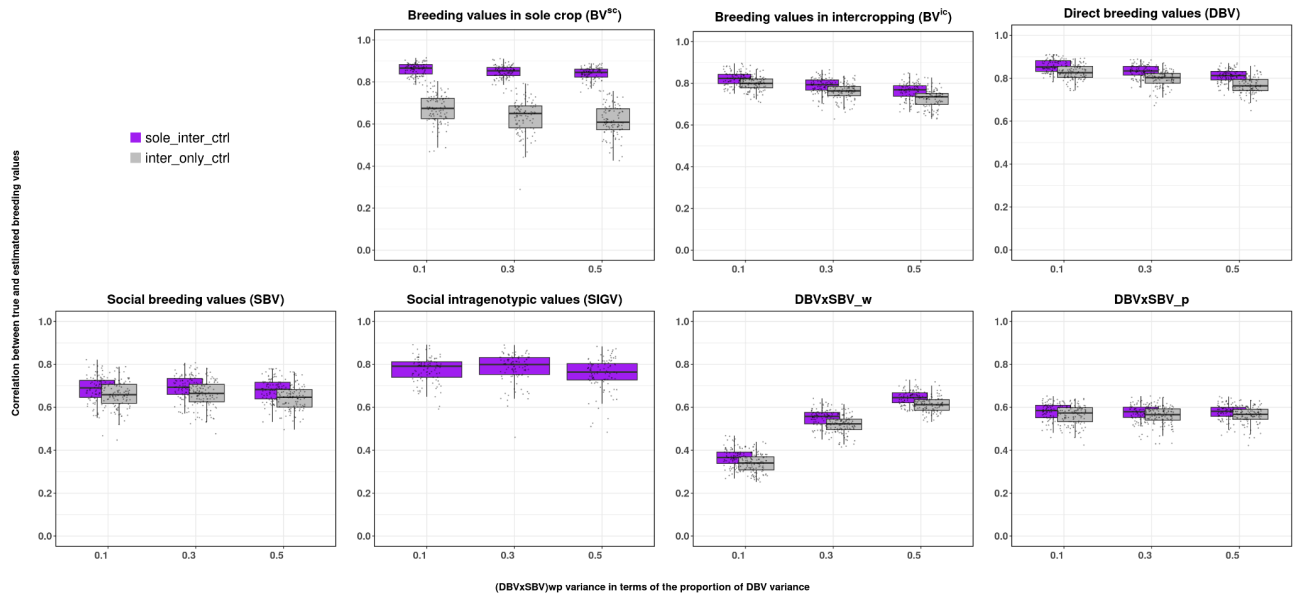
Supplementary Figure 10: Estimated genetic variance (y-axis) per targeted cropping system (x-axis; sole cropping, SC, or intercropping, IC), across the three experimental designs (sole_only, sole_inter_50 and inter_only). Results are shown across varying signs and magnitudes of $\text{cor}(\text{DBV}, \text{SBV})$: -0.8 (A), -0.3 (B), 0 (C), 0.3 (D) and 0.8 (E). Other (co)variance parameters were fixed as in figure 6: $\text{var}(\text{SBV}) = 20\% \text{ var}(\text{DBV})$ and $\text{var}(\text{SIGV}) = 10\% \text{ var}(\text{DBV})$. Error bars represent standard errors of the mean across replicates.

Supplementary Figure 11: Influence of the social genetic variance on genotypes selection



Supplementary Figure 11: Proportion of top-performing genotypes selected for sole cropping and intercropping systems across varying levels of social intragenotypic value variance and $BV^{SC}-BV^{IC}$ correlation. Results are shown for different levels of SBV variance expressed as a proportion of DBV variance (**A** = 10%; **B** = 30%; **C** = 50%), $\text{cor}(\text{DBV}, \text{SBV})$ being fixed at -0.8. Error bars represent standard errors across replicates. Tukey's HSD test was performed separately within each SIGV variance proportion; groups that do not share a common letter differ significantly ($\alpha = 0.05$). The x-axis indicates the cropping system under which selection was applied, where “sole” refers to selection for a sole cropping system and “mix” to selection for an intercropping system. Within each σ_{SIGV}^2 scenario, three selection strategies—sole_only, inter_only, and sole_inter_50—are compared.

Supplementary Figure 12: Accuracy of breeding values with $(\text{DBV} \times \text{SBV})^{\text{IC}}$



Supplementary Figure 12: Pearson correlations between true and estimated breeding values for BV^{SC} , BV^{IC} , DBV, SBV, SIGV, interaction effects ($\text{DBV} \times \text{SBV}_w$, which is the $(\text{DBV} \times \text{SBV})^{(\text{ft})}$) on focal (wheat) performance and interaction effects ($\text{DBV} \times \text{SBV}_p$, which is the $(\text{DBV} \times \text{SBV})^{(\text{tf})}$) on tester (pea) performance plotted on the y-axis. Simulation scenarios with increasing importance of the $\text{DBV} \times \text{SBV}_w$ are on the x-axis. $\text{DBV} \times \text{SBV}_w$ is expressed as the ratio $\text{var}(\text{DBV} \times \text{SBV}_w) / \text{var}(\text{DBV})$. Simulation parameters were $\text{var}(\text{SBV}) = 20\% \text{var}(\text{DBV})$, $\text{var}(\text{SIGV}) = 50\% \text{var}(\text{DBV})$, $\text{cor}(\text{DBV}, \text{SBV}) = -0.8$, $\text{cor}(\text{DBV} \times \text{SBV}_w, \text{DBV} \times \text{SBV}_p) = 0$, and $\text{var}(\text{DBV} \times \text{SBV}_p) = 30\% \text{var}(\text{DBV}_p)$. For the *inter_only* design, the true breeding value on solecrop (BV^{SC}) was compared only with the estimated direct breeding value because SIGV was not estimable under this design. The genetic similarity between mixtures is assumed to be identity $\text{K}^{(\text{ft}, \text{IC})} = \text{Id}$.